

SAHIL KAKADIYA

XR Developer | Unity Engineer | UI/UX Designer

Heilbronn, Germany | +49 155 66092668 | sahilkakadiya010@gmail.com

linkedin.com/in/sahil-kakadiya02 | github.com/SahilKakadiya2872 | sahil-vr-portfolio.vercel.app

PROFESSIONAL SUMMARY

Unity and XR developer with hands-on experience building immersive VR applications for Meta Quest 3, backed by a professional UI/UX design background. Strong in cross-engine asset pipelines (Unreal to Unity), physics-based interaction systems, and iterative on-device testing, with academic grounding in Mixed Reality, Realtime 3D Engines, and usability research. Currently completing an M.Sc. Software Engineering and seeking a working student, thesis, or internship role where VR engineering meets user-centred design. Unity VR Development certified by Unity Technologies.

TECHNICAL SKILLS

XR / Game Development: Unity (2D/3D/VR), Meta XR Interaction Toolkit, XR Interaction Toolkit, OpenXR, Shader Graph, Unreal Engine, Blender

Programming: C#, C++, Python, JavaScript, TypeScript, PHP, SQL

Web / Frontend: HTML, CSS, React, Next.js, Tailwind CSS

Design & UX: Figma, Adobe XD, Photoshop, Illustrator, InDesign, Prototyping, User Research, Usability Testing

Tools & Platforms: Git, GitHub, Meta Quest 3, Visual Studio, WordPress, Vercel

PROFESSIONAL EXPERIENCE

VR Development Intern

Aug 2025 – Feb 2026

UnityLab, Hochschule Heilbronn | Heilbronn, Germany

- Built an immersive VR garden and library scene for the TURTLE project, a university-wide VR learning platform deployed on Meta Quest 3 using Unity 6 and Meta XR Interaction Toolkit.
- Resolved cross-engine asset compatibility by manually extracting 3D models, textures, and materials from Unreal Engine and re-integrating them into Unity with full material remapping.
- Implemented a VR player rig (OVR Rig), teleportation locomotion, grabbable objects, and snap interaction systems, validated through iterative on-device testing on Quest 3 hardware.
- Diagnosed and fixed physics-based interaction issues including collider misconfigurations and snap-point misalignment across multiple interaction zones.
- Collaborated via GitHub within an 18-member cross-functional team, managing merge conflicts and maintaining scene consistency across branches.

UI/UX Designer

Jan 2023 – Sep 2023

Hevin Technoweb LLP | Surat, India

- Designed the complete UI for two of the company's product apps published on the Google Play Store, covering brand identity, navigation, and full screen flows.
- Created wireframes and interactive prototypes in Figma and Adobe XD across web and mobile projects, translating requirements into production-ready designs.
- Conducted user research and moderated usability tests, identifying friction points and iterating to improve task completion.
- Built responsive interfaces in HTML, CSS, and JavaScript, and produced branding assets in Illustrator and Photoshop.

PROJECTS

TURTLE VR – Hoffmann Experience

2025 – 2026

Unity 6 | Meta Quest 3 | Meta XR Interaction Toolkit | C# | Unreal Engine

GitHub repository

- Developed the Hoffmann Experience scene within the TURTLE VR learning platform: an interactive garden and library environment for educational exploration on Meta Quest 3.
- Handled end-to-end cross-engine asset migration from Unreal Engine to Unity, including mesh extraction, texture rebinding, and material shader conversion.

Audi VR Driving Simulation

2024

Mixed Reality Lab, Hochschule Heilbronn

Unity | VR | In-vehicle hardware rig

- Contributed debugging and testing to a multi-year VR driving simulation operated from a physical Audi vehicle connected to a VR headset, developed by the lab's thesis and doctoral researchers.
- Validated interaction and rendering behaviour across live in-vehicle VR sessions, identifying and reporting defects to improve stability.

Ninja Survival – 3D Game

2024

Unity | C# | Blender | Mixamo | Terrain Tools

GitHub repository · Gameplay video

- Built a third-person 3D survival prototype with dinosaur NPCs that play randomised Mixamo animations (idle, roar, eat) through Unity animation controllers, plus player run and jump movement and a boundary-based safe-zone mechanic.
- Designed the forest environment using Unity Terrain Tools, Blender, and custom lighting and texturing in Photoshop.

Yale School of Art – Usability Study

2024

Usability Evaluation & Testing, M.Sc. | Team of 8

Heuristic Evaluation | Moderated User Testing | Miro

- Ran a structured usability evaluation of the Yale School of Art website: heuristic analysis against Nielsen's ten principles, persona development, and four task-based test scenarios.
- Conducted five moderated user tests, measured success, struggle, and failure rates, compared results against the heuristic review, and prioritised recommendations by severity.

VR Room Experience

2025

Unity | XR Interaction Toolkit | OpenXR | C#

GitHub repository

- Built an interactive VR room with teleportation, grabbable objects, and physics-based interactions, completed as part of the official Unity VR Development certification pathway.

Wedding Villa – Web Application

2022

PHP | MySQL | JavaScript | Figma

- Full-stack wedding-planning web application built as Bachelor's capstone, with a vendor database and interfaces designed in Figma.

EDUCATION

M.Sc. Software Engineering

Mar 2024 – Present

Hochschule Heilbronn | Heilbronn, Germany

GPA: 2.1 (German scale) | Coursework: Mixed Reality Lab, Realtime 3D Engines, Cloud Computing, DevOps & SecOps, Advanced Software Architecture, Usability Evaluation

B.C.A. (Bachelor of Computer Applications)

Sep 2019 – Apr 2022

Rabindranath Tagore University | India

CERTIFICATIONS

Unity VR Development – Unity Technologies Sep 2025

Unity Essentials – Unity Technologies Oct 2024

GIM & Web Designing – Red & White Multimedia Educations 2022

LANGUAGES

English: C1 (Professional Working Proficiency) German: B1 (Intermediate)